Berliner Hochschule für Technik

FB III – M. Sc. Geoinformation

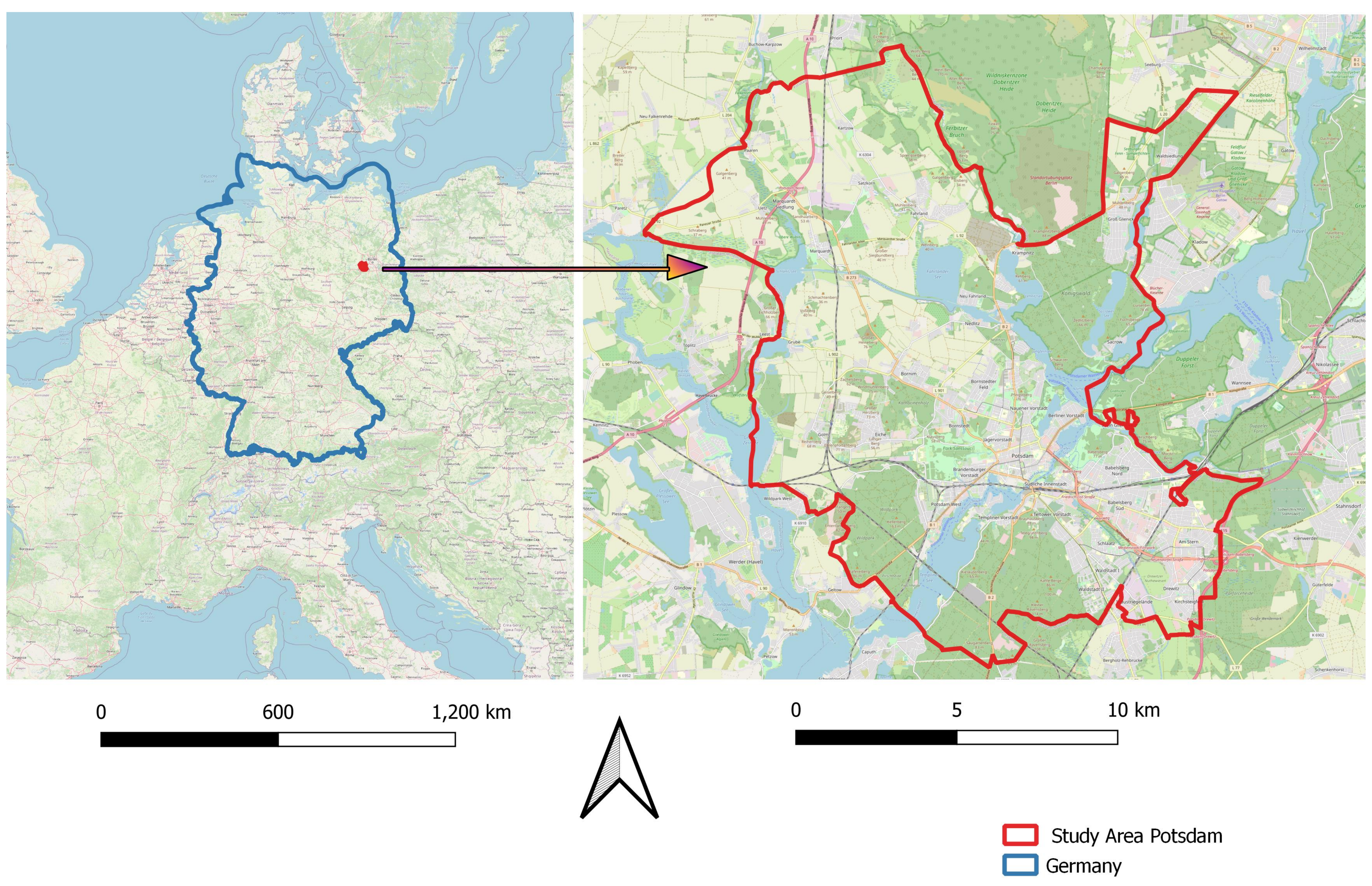
Classification of Remote Sensing Data

A study on the Classification of remote sensing data using Spatial Modeler in Erdas Imagine and Random Forest Algorithm

Objective

The objective of this study is to classify the sentinel- 2 satellite image data using Spatial Modeler feature in Erdas Imagine and machine learning algorithm "Random Forests" using supervised classification method. The accuracy of the classification results is then determined visually, and the obtained results are compared with the Corine Land Cover -2018 dataset which itself is a classified data covering different land cover classes.

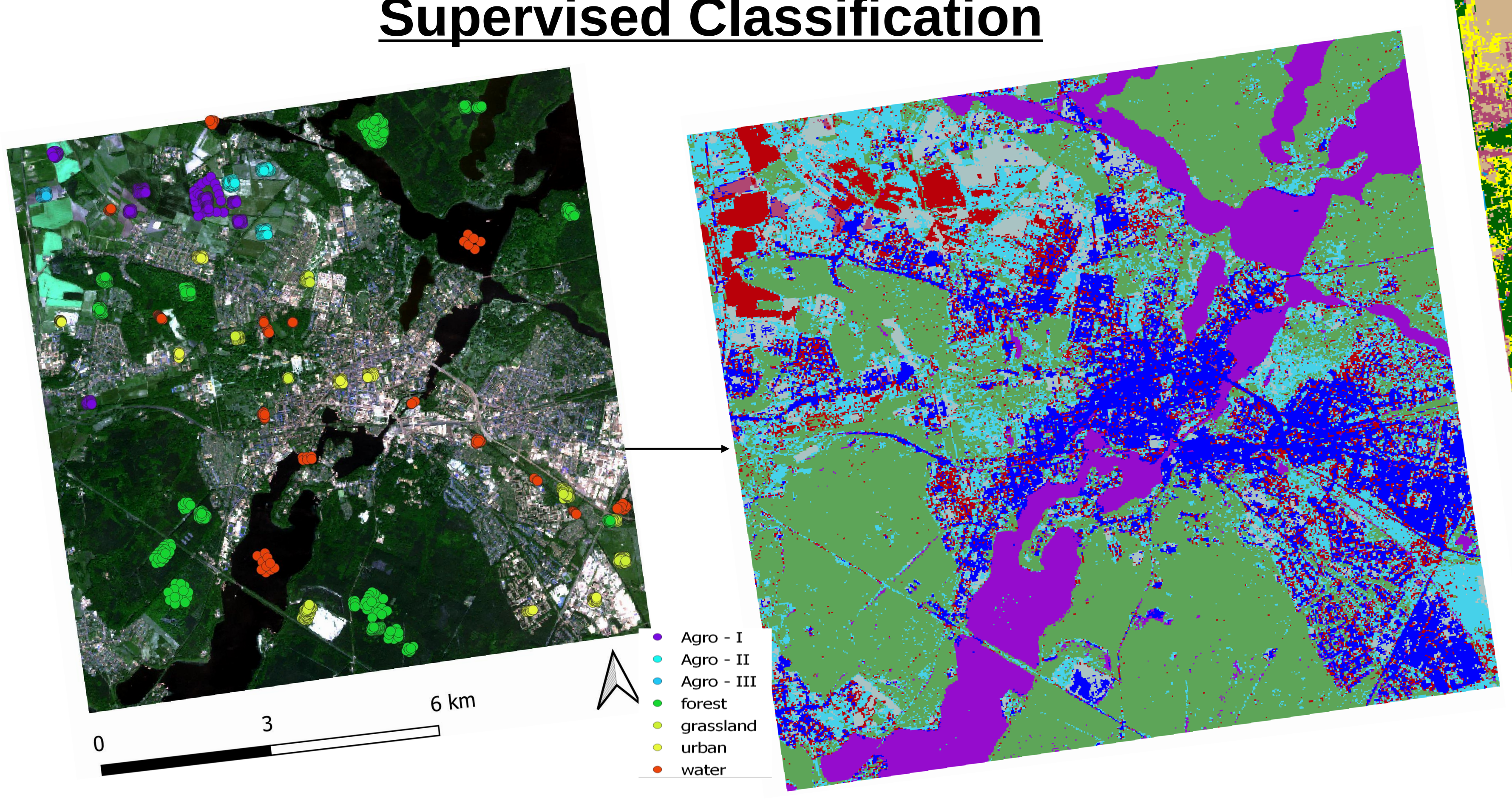
Area of Study



Schematic workflow!!!!

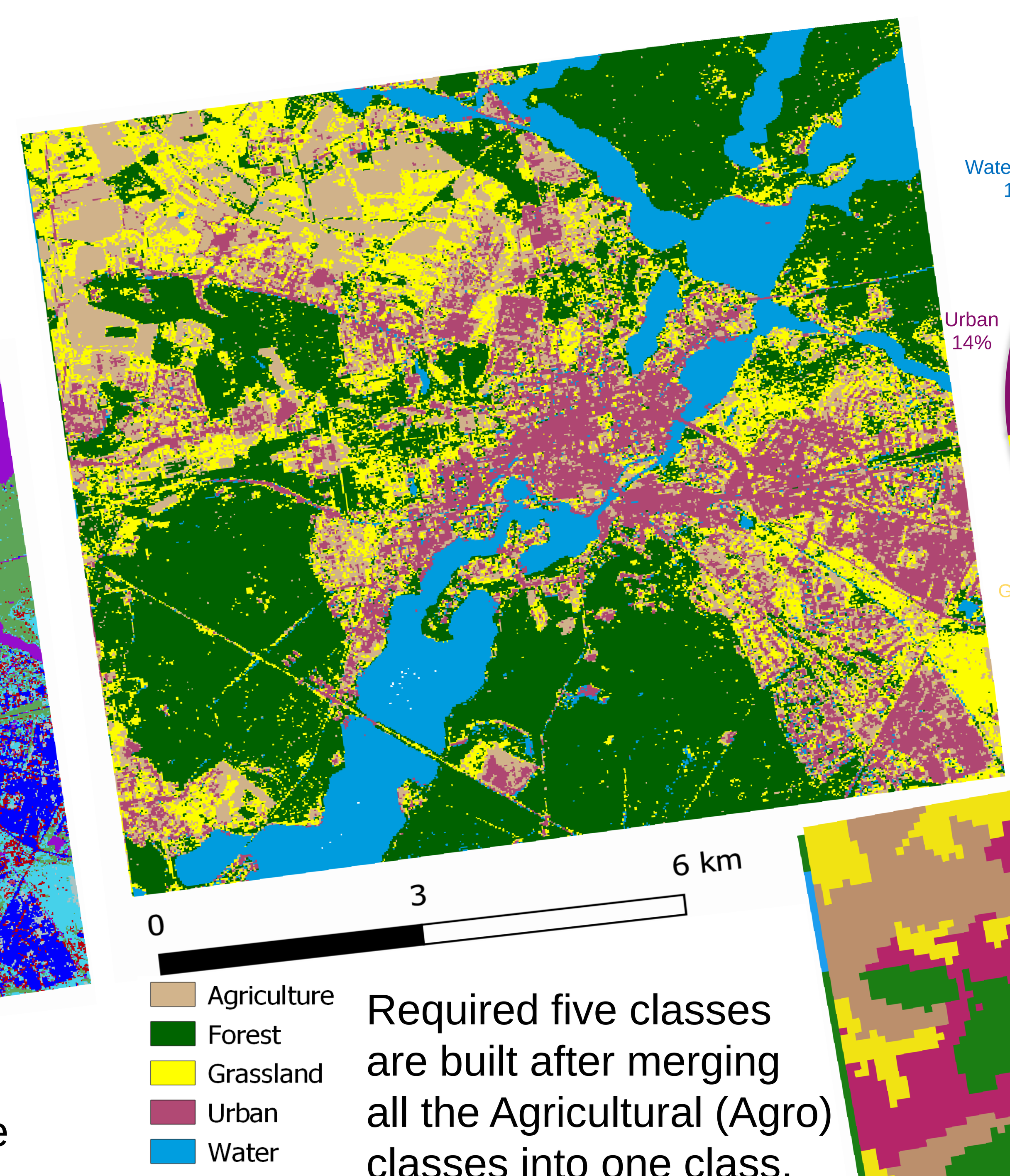
Study Area → Sentinel-2 Images → Image Processing → Training sample points Preparation → Training the Machine Learning Algorithm → Classifyng using Machine Intellect → Accuracy Assessment → Stratified Random Sampling → Asseessment Report

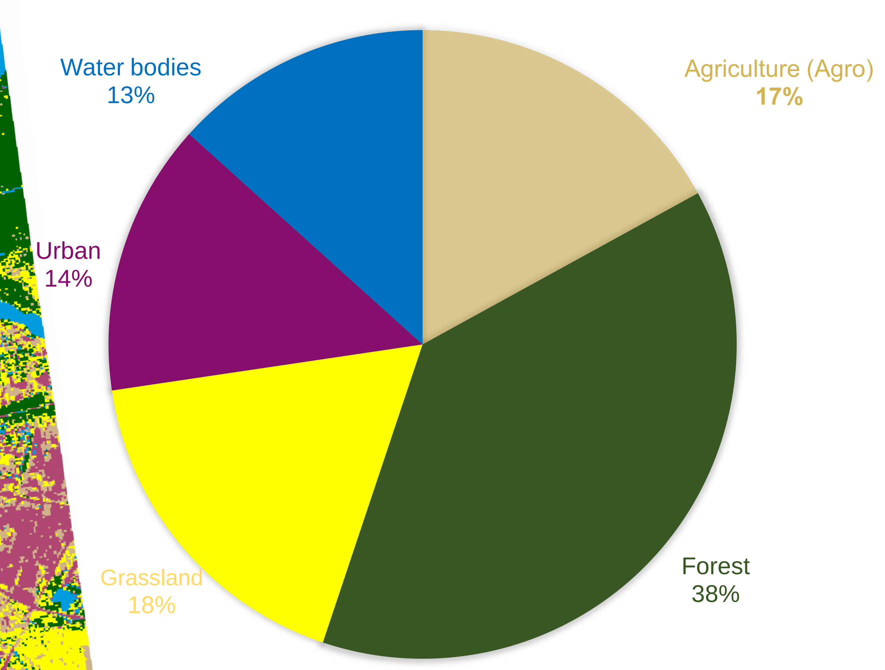
Supervised Classification



To classify the Raster Image 8 different Training classes datasets are created in the form of vector dataset.

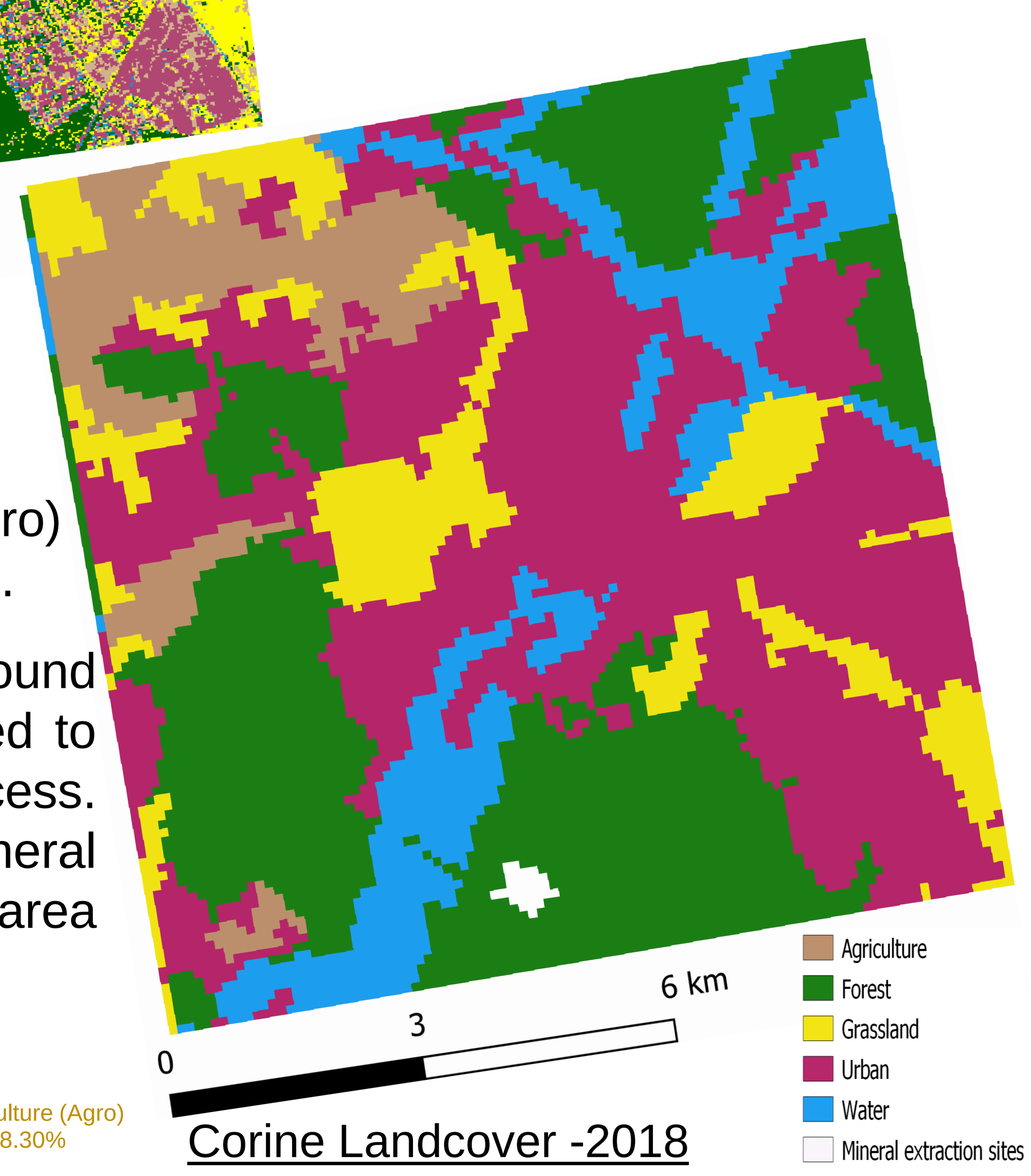
Supervised classification entails the development of spectral signatures by the user for the features such as urban, water, forest, etc. Then the processing software compares each pixel in the image to the cover type similar to its signature. The red, gray, and maroon color bands depict the agricultural land, whereas green illustrates the forest areas. The agricultural field was assigned during the classification process in three different classes due to its different spectral properties. The urban area is represented by blue, and the turquoise shows the grassland area. The water body is easily recognizable in its purple color.



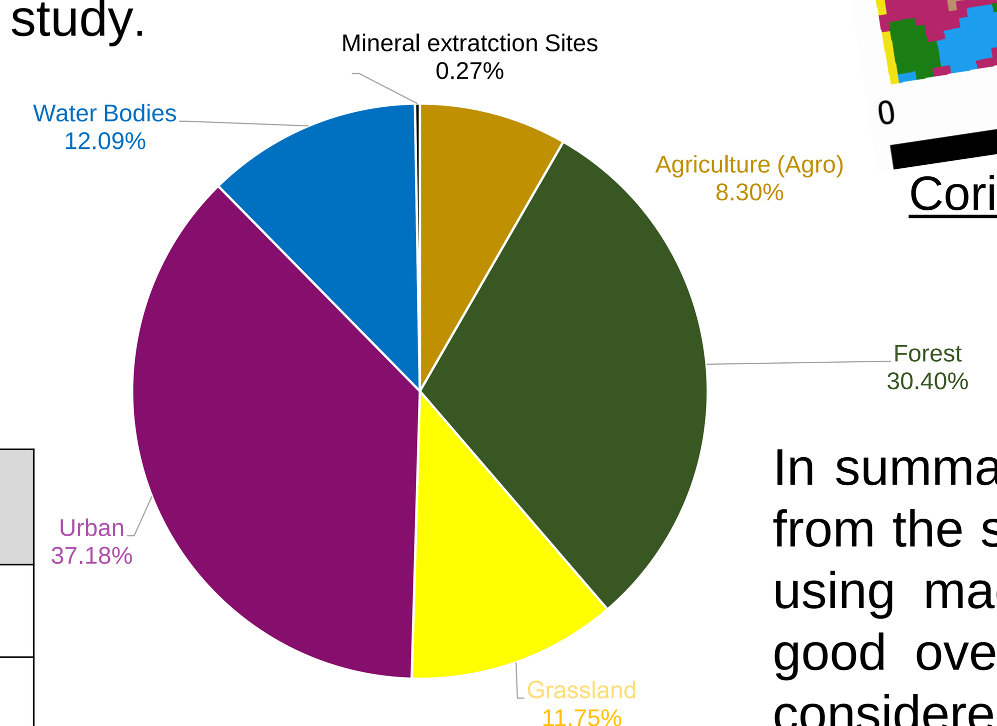


Required five classes are built after merging all the Agricultural (Agro) classes into one class.

A new Mineral extraction was found during the process, which needed to be considered during the process. Hence a new class as Mineral extraction sites was given in the area of study.



Corine Landcover -2018



Accuracy Assessment

S. No.	Classified Data	Agriculture (Agro)	Forest	Grassland	Urban	Water	Row Total
1	Agriculture (Agro)	9	1	7	9	0	26
2	Forest	1	52	1	0	0	54
3	Grassland	6	4	15	2	0	27
4	Urban	1	1	2	18	0	22
5	Water	0	1	0	0	20	21
	Total	17	59	25	29	20	150

The highlighted diagonal elements represents the number of correctly classified points.

Class	Producer's Accuracy	User's Accuracy
Agriculture	52.94%	34.52%
Forest	88.14%	96.30
Grassland	60.07%	55.56%
Urban	62.07%	81.82%
Water	100.00%	95.24%
Overall Accuracy	76%	

S. No.	Classified Data	Agriculture (Agro)	Forest	Grassland	Urban	Water	Row Total
1	Agriculture (Agro)	8	0	5	17	0	30
2	Forest	2	14	5	9	0	30
3	Grassland	5	3	9	13	0	30
4	Urban	0	0	4	26	0	30
5	Water	0	2	0	2	26	30
	Total	15	15	23	67	26	150

Class	Producer's Accuracy	User's Accuracy
Agriculture	53.33%	26.67
Forest	73.68%	46.67
Grassland	39.13%	30.00%
Urban	38.81%	86.67%
Water	100.00%	86.67%
Overall Accuracy	55.33%	

In summary, it can be said that the result from the supervised classification process using machine learning algorithm has a good overall accuracy and thus can be considered as a suitable classification method. The comparison with the Corine landcover- 2018 was not suitable though as it has a lower accuracy because of various reasons such as lower geometric resolutions and strongly generalised datasets.

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